DECADE (BCD) RIPPLE COUNTER

MOD-10 ⇒ States = 10 = 2^n

Max Count = 10 − 1

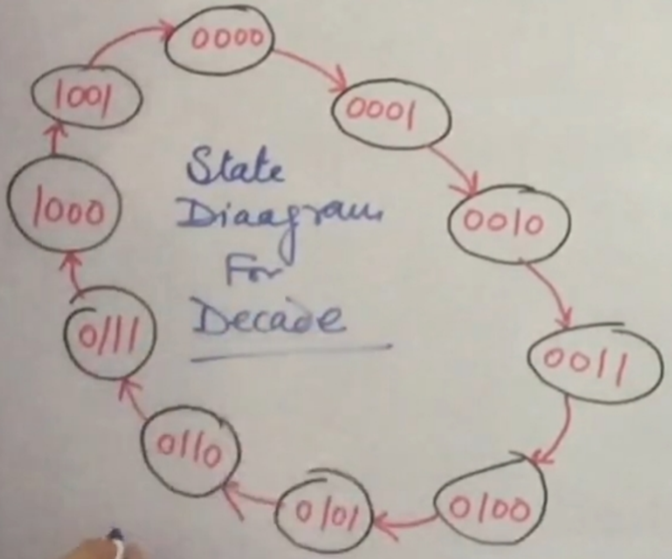
= $2^n - 1$

= 9

BCD

0 ⟶ 9

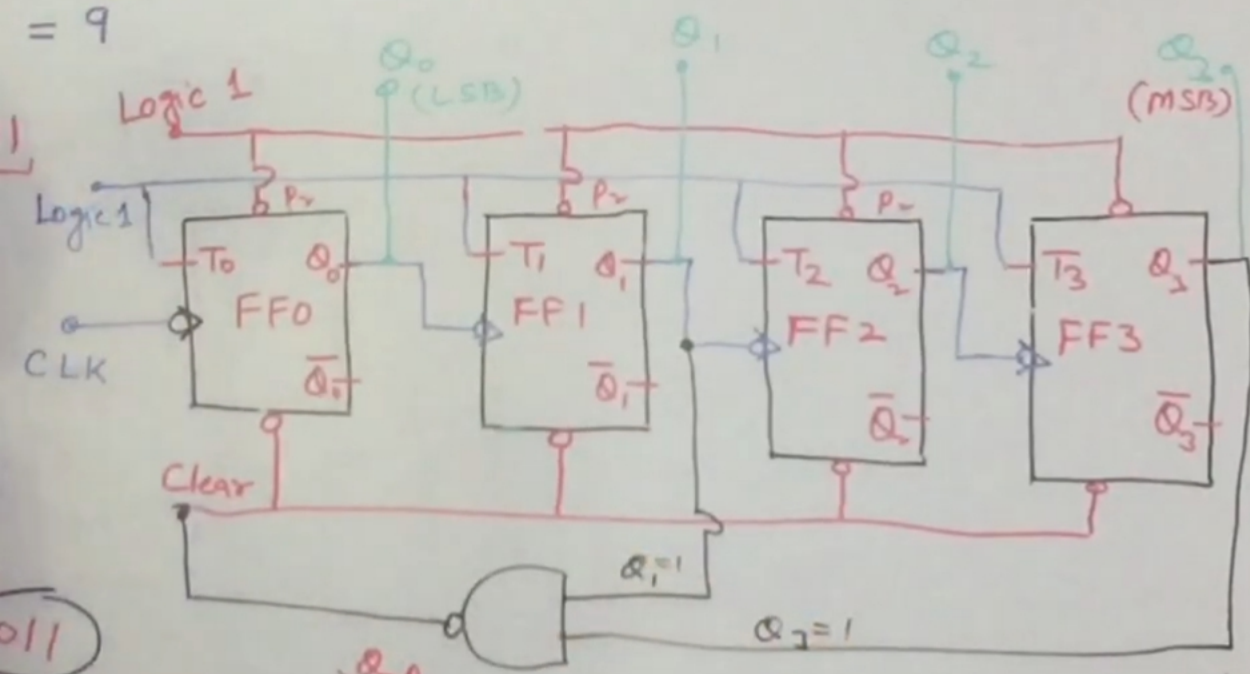
0000 ⟶ 1001



DECADE (BCD) RIPPLE COUNTER

$= 10 = 2^n$
$= 10 - 1$
$= 2^n - 1$
$= 9$

$\left.\begin{matrix} Cr \\ Pr \end{matrix}\right\}$ = 1 Normal oper.
= 0] Active



Dec	Q_3	Q_2	Q_1	Q_0	Cr
0	0	0	0	0	1
1	0	0	0	1	1
2	0	0	1	0	1
3	0	0	1	1	1
4	0	1	0	0	1
5	0	1	0	1	1
6	0	1	1	0	1
7	0	1	1	1	1
8	1	0	0	0	1
9	1	0	0	1	1
10	$\boxed{1}$	0	$\boxed{1}$	0	0
11	1	0	1	1	X
12	1	1	0	0	X
13	1	1	0	1	X
14	1	1	1	0	X
15	1	1	1	1	X

Q_1Q_0 \ Q_3Q_2	00	01	11	10
00	1	1	X	1
01	1	1	X	1
11	1	1	X	X
10	1	1	X	0

$$Cr = \overline{Q}_3 + \overline{Q}_1 = \overline{Q_3 \cdot Q_1}$$